

Statistics

2025 Lectures

Part 3 - Multivariate Distributions and Transformations

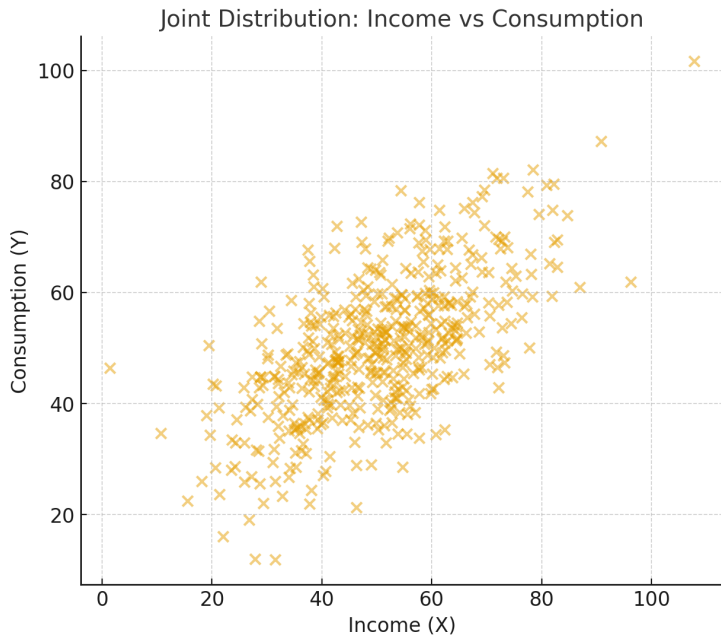
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Recommended reading (week 2)

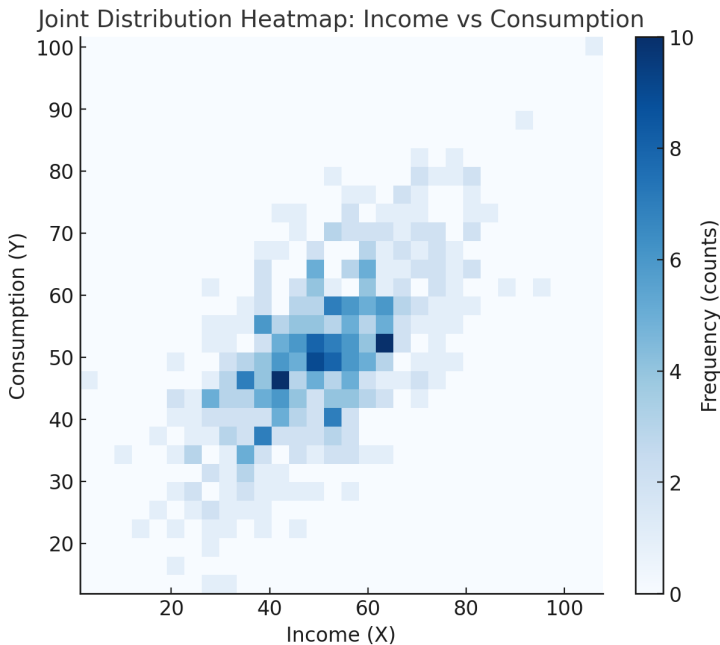
For the week 2 lectures, please, read

- sections 6.1, 6.2, 6.3, 6.4 from Bartoszynski and Niewiadomska-Bugaj (33 pages)
- sections 2.4, 2.5 from Mittelhammer (16 pages)

Motivation: Income vs Consumption



Motivation: Income vs Consumption



Example of pair of random variables

- In many practical situations, we often need to work with multiple random variables, which may be dependent on each other.
- The analysis of random variables can be extended to the case of multiple random variables, i.e., to the analysis of **random vectors**, focusing on their joint effects.

Example 20: Consider an experiment involving three tosses of a coin, where X represents the “number of heads in all three tosses” and Y represents the “number of tails in the last two tosses”.

- To simplify the notation, we write $P(X = x, Y = y)$ instead of $P(\{X = x\} \cap \{Y = y\})$ or $P(\{s \in S | X(s) = x\} \cap \{s \in S | Y(s) = y\})$.

Joint discrete and continuous distribution

Definition 9: We say that the pair (X, Y) of random variables has a discrete distribution if there exist finite or countable sets A and B such that

$$P((X, Y) \in A \times B) = \sum_{x \in A, y \in B} P(X = x, Y = y) = 1.$$

Example 20 (cont.): Clearly, we have $A = \{0, 1, 2, 3\}$ and $B = \{0, 1, 2\}$.

Definition 10: Random variables (X, Y) are jointly continuous if there exists a function $f(x, y)$, $x, y \in \mathbb{R}$, such that for every

$$C = \{(x, y) | a \leq x \leq b, c \leq y \leq d\},$$

where $-\infty \leq a < b \leq \infty$, $-\infty \leq c < d \leq \infty$, we have

$$P((X, Y) \in C) = \iint_{(x, y) \in C} f(x, y) \mathrm{d}(x, y).$$

Fubini theorem for integral

Theorem 8c: (Fubini for Lebesgue integral)

Suppose that $f : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$ and $A \in \mathcal{B}(\mathbb{R}^m)$, $B \in \mathcal{B}(\mathbb{R}^n)$. If

$$\iint_{A \times B} |f(x, y)| d(x, y) < \infty,$$

i.e. f is absolutely integrable, then

$$\iint_{A \times B} f(x, y) d(x, y) = \int_A \left(\int_B f(x, y) dy \right) dx = \int_B \left(\int_A f(x, y) dx \right) dy.$$

- In Definition 10, f , referred to as the **joint density function** of (X, Y) , is nonnegative almost everywhere. Therefore, the double integral and both iterated integrals are either all infinite or all equal to the same finite value.

Fubini theorem for sums

Theorem 8d: (Fubini for sums)

Suppose that $a_{ij}, i, j \in \mathbb{N}$, is absolutely summable, i.e.

$$\sum_{i,j \in \mathbb{N}} |a_{ij}| < \infty.$$

Then

$$\sum_{i,j \in \mathbb{N}} a_{ij} = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

- We consider a_{ij} 's to be nonnegative since they represent probabilities (see Definition 9). Therefore, the double sum and both iterated sums are either all infinite or all equal to the same finite value.

Role of Fubini theorem and marginal density functions

- With the use of Fubini theorem

$$\begin{aligned} P((X, Y) \in C) &= \iint_{(x,y) \in C} f(x, y) \mathrm{d}(x, y) = \\ &= \int_a^b \left[\int_c^d f(x, y) \mathrm{d}y \right] \mathrm{d}x = \int_c^d \left[\int_a^b f(x, y) \mathrm{d}x \right] \mathrm{d}y. \end{aligned}$$

- In general

$$\iint_C f(x, y) \mathrm{d}(x, y) = \int_{C_2} \varphi(y) \mathrm{d}y = \int_{C_1} \psi(x) \mathrm{d}x,$$

where

$$C_1 = \{x | (x, y) \in C\}, C_2 = \{y | (x, y) \in C\}.$$

- We refer to φ and ψ as to the **marginal density functions** of (X, Y) .

Joint distribution function

Example 21: Let

$$f(x, y) = \begin{cases} cx(x + y) & x \geq 0, y \geq 0, x + y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find $P(X \leq 1/2)$.



Joint distribution function

Definition 11: Function $F_{X,Y}(x, y) = F(x, y) = P(X \leq x, Y \leq y)$ is called the **cumulative distribution function (cdf) of (X, Y)** .

- in discrete case $F(x, y) = \sum_{x_i \leq x, y_i \leq y} P(X = x_i, Y = y_i)$;
- in continuous case $F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) d(x, y)$.

Theorem 9: The cdf F has the following properties:

- i) $\lim_{x, y \rightarrow \infty} F(x, y) = 1$.
- ii) For every y , $F(\cdot, y)$ is nondecreasing and continuous from the right.
- iii) For every x , $F(x, \cdot)$ is nondecreasing and continuous from the right.
- iv) $\lim_{x \rightarrow -\infty} F(x, y) = 0 \quad \forall y$; $\lim_{y \rightarrow -\infty} F(x, y) = 0 \quad \forall x$.
- v) For all $x_1 < x_2$ and $y_1 < y_2$

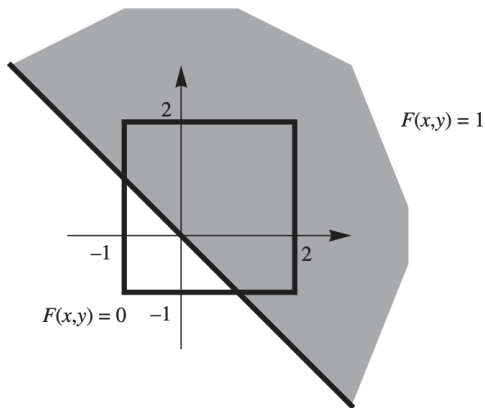
$$F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) \geq 0.$$

Joint distribution function

Example 22: Let F be defined by the formula

$$F(x, y) = \begin{cases} 1 & \text{for } x + y \geq 0; \\ 0 & \text{for } x + y < 0. \end{cases}$$

Consider e.g. $(x_1, x_2) = (-1, 2)$ and $(y_1, y_2) = (-1, 2)$.



Marginal distributions

- Given the bivariate distribution, we can recover both univariate marginal distributions of X and Y , but the reverse is not possible; knowing only the marginal distributions does not allow us to reconstruct the full bivariate distribution.
- discrete case: $A = \{x_1, x_2, \dots\}, B = \{y_1, y_2, \dots\}$ sets of values of X and Y

Let $p_{ij} = P(X = x_i, Y = y_j)$ then since
 $\{X = x_i\} = \cup_j (\{X = x_i, Y = y_j\})$

$$P(X = x_i) = \sum_j p_{ij} := p_{i\cdot}, \quad P(Y = y_j) = \sum_i p_{ij} := p_{\cdot j}.$$

- continuous case:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

are called marginal densities of X and Y .

Independence of marginal rv's

Definition 12: We say that random variables X and Y are **independent** if events $\{X \in A\}, \{Y \in B\}$ are independent for all sets A, B . The latter holds if and only if

$$F(x, y) = F_X(x) \cdot F_Y(y) \quad \text{for every } x \text{ and } y.$$

■ discrete case: $p_{ij} = p_{i\cdot} \cdot p_{\cdot j} \quad \forall i, j$

■ continuous case: $f(x, y) = f_X(x) \cdot f_Y(y)$ for almost all x and y

Example 23:

$$f(x, y) = \begin{cases} cxy^2, & x \geq 0, y \geq 0, x + y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Determine if X and Y are independent.

Application of independence

- The formula in Definition 12 is useful not only for determining whether two variables are independent, but it can also be used to find the joint distribution of independent variables X and Y .

Example 24: A man makes two attempts to achieve a goal. His performance X during the first attempt, measured on a scale from 0 to 1, is a random variable with the following density function

$$f_1(x) = 12x^2(1 - x), \quad 0 \leq x \leq 1.$$

His performance Y during the second attempt is independent of X , and its density function is given by

$$f_2(y) = 6y(1 - y), \quad 0 \leq y \leq 1.$$

What is the probability that the man exceeds the level of 0.75 in the better of the two attempts?

General case

- Consider a general **random vector** $X = (X_1, \dots, X_n)^\top$, which is a vector comprising the random variables $X_1, \dots, X_n$.
- In the discrete case, the distribution of X is determined by

$$P_X(x) = P(X = x) = P(X_1 = x_1, \dots, X_n = x_n),$$

where $x = (x_1, \dots, x_n)^\top$ and $\sum_x P_X(x) = 1$.

- In the continuous case, the joint density of X is a nonnegative function f_X mapping $\mathbb{R}^n$ to $\mathbb{R}$ such that

$$P(X \in Q) = \int \cdots \int_{x \in Q} f(x_1, \dots, x_n) \mathbf{d}(x_1 \dots x_n)$$

with

$$\int \cdots \int_{\mathbb{R}^n} f(x_1, \dots, x_n) \mathbf{d}(x_1 \dots x_n) = 1.$$

General marginal distributions

- The concepts of marginal distributions and densities remain the same; however, they can also be multivariate.

Definition 13: Let $X = (Y, Z)$, Y and Z random vectors.

In the discrete case the **marginal distribution of Y** is given by

$$p_Y(y) = \sum_z P(Y = y, Z = z).$$

In the continuous case, the **marginal density of Y** is given by

$$f_Y(y) = \int \cdots \int_z f(y, z) dz.$$

Example 25: Consider an experiment with 3 outcomes A, B and C with probabilities α, β and γ , respectively, $\alpha + \beta + \gamma = 1$, and consider X_1 as counts of A among n repetitions and X_2 as counts of B among n repetitions. Find (marginal) distributions of X_1 and X_2 .

Multivariate Transformations

- Determining the densities of bivariate or multivariate transformations is generally considered to be challenging.
- Follow the steps:
 - a) To a transformation function φ choose a “companion function” η such that

$$z = \varphi(x, y)$$

$$w = \eta(x, y)$$

can be solved as $x = \alpha(z, w)$ and $y = \beta(z, w)$.

- b) Determine the image D of the support C of density f in (z, w) -plane under the transformation.

Multivariate Transformations

- c) Find the Jacobian of the transformation, i.e. the determinant

$$J = \begin{vmatrix} \frac{\partial \alpha}{\partial z} & \frac{\partial \alpha}{\partial w} \\ \frac{\partial \beta}{\partial z} & \frac{\partial \beta}{\partial w} \end{vmatrix}$$

- d) Determine the joint density of (Z, W) ,
 $Z = \varphi(X, Y)$, $W = \eta(X, Y)$ by

$$g(z, w) = \begin{cases} f(\alpha(z, w), \beta(z, w))|J| & \text{for } (z, w) \in D, \\ 0 & \text{otherwise.} \end{cases}$$

- e) Compute the density of Z as the marginal density of (Z, W)

$$g_Z(z) = \int_{-\infty}^{\infty} g(z, w) \mathrm{d}w = \int_{D_z} f(\alpha(z, w), \beta(z, w))|J| \mathrm{d}w,$$

where $D_z = \{w | (z, w) \in D\}$.

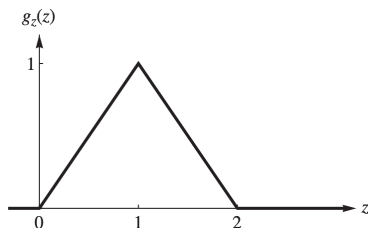
Multivariate Transformations

Example 26: (sum of random variables)

Determine the general density function for the sum $Z = X + Y$ of two continuous random variables X and Y .

Example 27: (triangular distribution)

Determine the distribution of $Z = X + Y$, when $X, Y \sim U(0, 1)$ independent.



Example 28: (product of random variables)

Determine the general density function for the product XY of two continuous random variables X and Y .

Recommended reading (week 3)

For the (next) week 3 lecture, please, read

- sections 7.1, 7.2, 7.4, 7.5, 7.6 from Bartoszynski and Niewiadomska-Bugaj (33 pages)
- sections 3.1, 3.2, 3.3, 3.4 from Mittelhammer (15 pages)